MANASVI SAXENA

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SUMMARY

Research scientist and entrepreneur with a PhD in Computer Science and extensive experience in developing real-world software systems with formal-methods-based correctness guarantees. Proven track record of transitioning academic research into production-scale systems, including direct experience with Amazon's S3 Automated Reasoning Group.

EDUCATION

Ph.D. Computer Science, University of Illinois Urbana-Champaign Dec 2024

- Advisor - Prof. Grigore Roşu
- Thesis - A Semantics-First Approach to Safe Guidelines-Based Clinical Decision Support

M.S. Computer Science, University of Illinois Urbana-Champaign May 2018

- Relevant Coursework - Formal Methods of Software Development, Runtime Verification, Logic in Computer Science, Computer and Cryptocurrency Security

B.S. Computer Science, University of Illinois Urbana-Champaign May 2015

- Relevant Coursework - Programming Languages and Compilers, Fundamental Algorithms, Theory of Computation, Computer Networking, Artificial Intelligence

EXPERIENCE

Satyax Inc., Urbana, Illinois January 2025 - Present
Founder & CEO *Urbana, IL*

- Building a platform for trustworthy advanced clinical decision support that integrates intelligence with formal-methods-based mathematical safety guarantees. Our software assists with acute conditions such as sepsis, representing the first formally verified comprehensive clinical decision support system with mathematical proofs of desired correctness properties.
- Translating doctoral research on a formal-semantics-driven approach to encoding medical knowledge in a computable yet clinician-comprehensible form, enabling the development of intelligent yet safe clinical decision support systems that foster greater clinician confidence through transparency, addressing concerns of safety and opaqueness associated with black-box approaches.
- Driving early-stage product development, customer discovery, and grant strategy to lay the foundation for commercialization.

S3 Automated Reasoning Group, Amazon Web Services (AWS) Summer 2021 & 2020
Applied Scientist Intern

- Developed techniques and tooling for evaluating compliance of highly concurrent, distributed code against relevant properties expressed as RV-Monitor specifications (`https://github.com/runtimeverification/rv-monitor`) utilizing the existing test suite.
- Optimized RV-Monitor, an academic tool, to handle millions of lines of code, and enabled support for expressing properties relevant to concurrent systems, such as read after write consistency and serializability efficiently, enabling RV-Monitor to handle the scale of massively parallel and highly distributed systems like S3.

Formal Systems Laboratory, University of Illinois May 2016 - December 2024
Research Assistant *Urbana, IL*

- Developed the MediK Framework, a DSL for encoding medical guidelines using state machines with strong correctness guarantees (`https://github.com/medik-framework`).
- Maintained RV-Monitor, a language independent monitoring framework that supports rich formalisms for expressing properties.
- Contributed to KEVM, formalizing the semantics of the Ethereum Virtual Machine (EVM) in the K Framework (`http://github.com/runtimeverification/evm-semantics`).

Runtime Verification Inc., Urbana, Illinois July 2015 - May 2016
Formal Verification Engineer *Urbana, IL*

- Worked on a team developing a dynamic analysis tool for finding undefined behavior in C programs, using formal semantics of C.
- Worked on improving tool's performance, and evaluated its applications on large codebases.

PROJECTS

The MediK Framework - Trustworthy Intelligent Decision Support

- *Domain-Specific Language* for encoding medical guidelines as verifiable state machines.
- *Safety guarantees* through mathematical proofs while maintaining clinical interpretability.
- *Comprehensive approach* combines formal reasoning with clinician-validated clinical knowledge.
- *Repository:* `https://github.com/medik-framework`

RV-Monitor

- *Core team member* maintaining language-agnostic monitor framework.
- *Optimized for enterprise-scale systems* with *rich formalisms* for expressing complex properties.
- *Repository:* `https://github.com/runtimeverification/rv-monitor`

KEVM - Semantics of the EVM in K

- *Formalization* of the Ethereum Virtual Machine (EVM) in the K Framework. Contributed to the semantics of the Ethereum ABI, as an extension to the original semantics.
- *Enables formal analysis* of smart contracts with high degree of automation.
- *Repository:* `https://github.com/runtimeverification/evm-semantics`

CORE COMPETENCIES

Automated Reasoning:	Formal Semantics, Static and Dynamic Analysis, Runtime Monitoring
Safety-critical Software:	Trustworthy AI-enabled Systems, Clinician-validated Decision Support
Programming:	Java, Haskell, K Framework, Scala, Python, C

PUBLICATIONS

- Manasvi Saxena. *A semantics-first approach to safe guidelines-based clinical decision support*. PhD thesis, University of Illinois Urbana-Champaign, December 2024. Available at `https://hdl.handle.net/2142/127206`
- Manasvi Saxena, Shuang Song, and Lui Sha. MediK: Towards Safe Guideline-based Clinical Decision Support. In *2023 Formal Methods in Computer-Aided Design (FMCAD)*, pages 306–317. IEEE, 2023
- Everett Hildenbrandt, Manasvi Saxena, Nishant Rodrigues, Xiaoran Zhu, Philip Daian, Dwight Guth, Brandon Moore, Daejun Park, Yi Zhang, Andrei Stefanescu, et al. KEVM: A complete formal semantics of the Ethereum Virtual Machine. In *2018 IEEE 31st Computer Security Foundations Symposium (CSF)*, pages 204–217. IEEE, 2018
- Dwight Guth, Chris Hathhorn, Manasvi Saxena, and Grigore Rosu. Rv-match: Practical semantics-based program analysis. In *Computer Aided Verification - 28th International Conference, CAV 2016, Toronto, ON, Canada, July 17-23, 2016, Proceedings, Part I*, volume 9779 of *LNCS*, pages 447–453. Springer, July 2016
- Daejun Park, Yi Zhang, Manasvi Saxena, Philip Daian, and Grigore Rosu. A formal verification tool for ethereum VM bytecode. In Gary T. Leavens, Alessandro Garcia, and Corina S. Pasareanu, editors, *Proceedings of the 2018 ACM Joint Meeting on European Software Engineering Conference and Symposium on the Foundations of Software Engineering, ESEC/SIGSOFT FSE 2018, Lake Buena Vista, FL, USA, November 04-09, 2018*, pages 912–915. ACM, 2018
- Shuang Song, Manasvi Saxena, Pei-Hsuan Tsai, and Lui Sha. Towards modular and formally-verifiable software architecture for clinical guidance systems. In *2023 IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, pages 4271–4276. IEEE, 2023
- Philip Daian, Dwight Guth, Chris Hathhorn, Yilong Li, Edgar Pek, Manasvi Saxena, Traian Florin Serbanuta, and Grigore Rosu. Runtime verification at work: A tutorial. In *Runtime Verification - 16th International Conference, RV 2016 Madrid, Spain, September 23-30, 2016, Proceedings*, volume 10012 of *Lecture Notes in Computer Science*, pages 46–67. Springer, September 2016